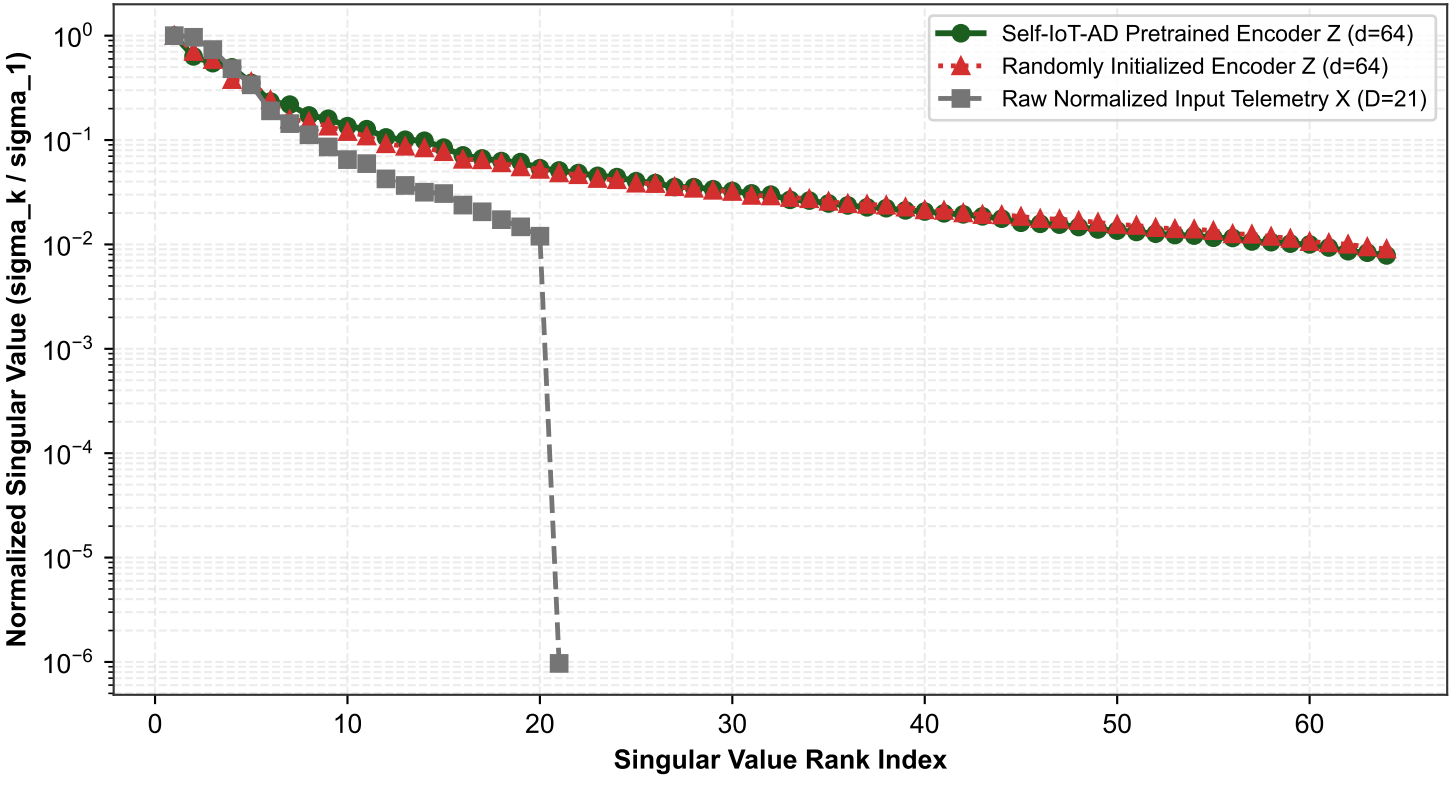
Figure 5: Singular Value Spectrum of Learned Latent Space (RQ1: Absence of Dimensional Collapse)
Self-IoT-AD Pretrained Encoder Z (d=64)
Randomly Initialized Encoder Z (d=64)
Raw Normalized Input Telemetry X (D=21)
Normalized Singular Value (sigma_k / sigma_1)
Singular Value Rank Index
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
10
20
30
40
50
60